Online Quiz System - Sprint 2

Team 73 |

Sprint 2 Server Execution & HTTP Request Logs

System Environment | Flask & SQLite

The Flask application server runs within an isolated macOS virtual environment (venv). Terminal execution confirms process cleanup on port 5000, debug mode activation, and consecutive real-time HTTP 200/302 logging across all candidate workflows and administrator operations.

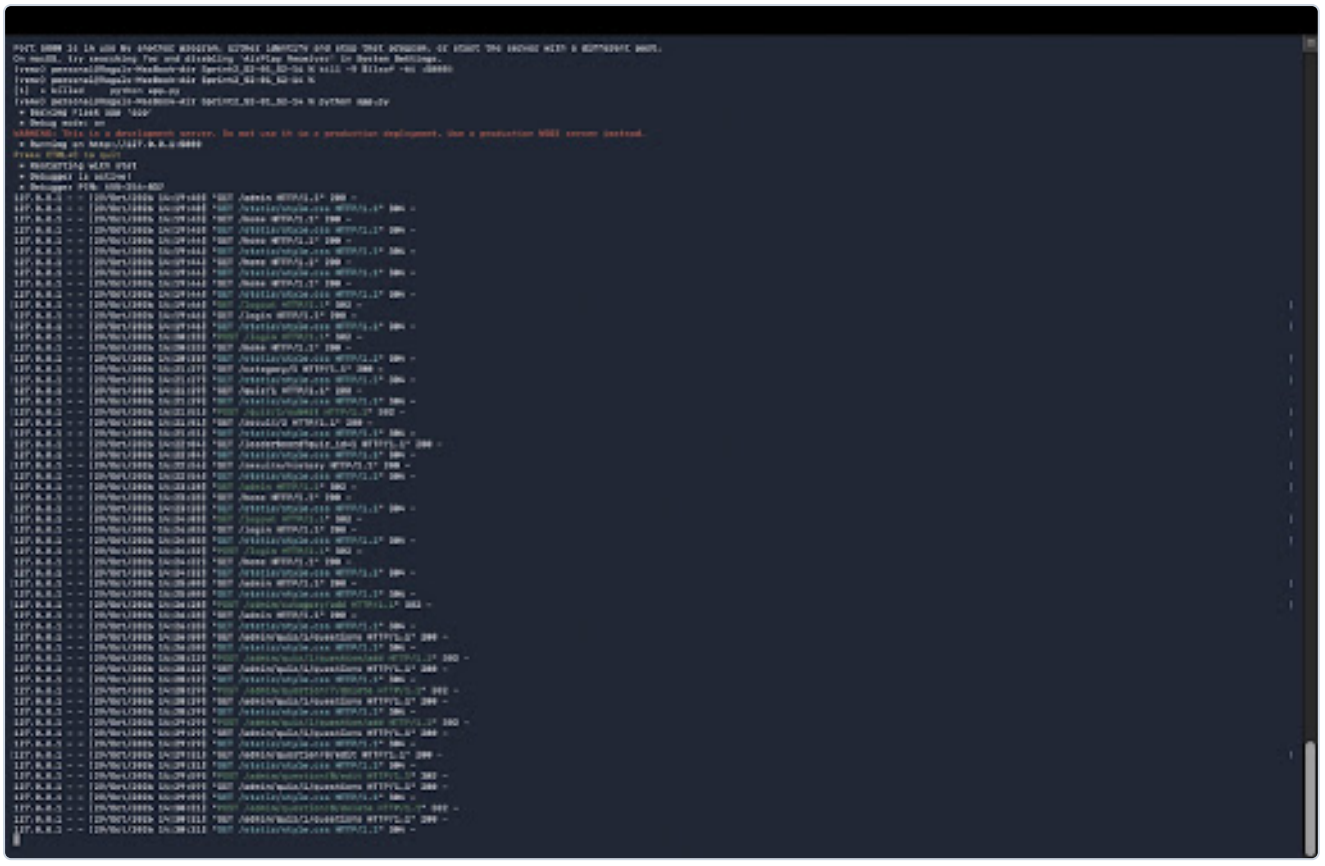


Figure 1.1: Live terminal logs displaying Flask server initialization and HTTP route traces from 14:19:40 to 14:30:31.

```
(venv) personal@Raguls-MacBook-Air Sprint2_S2-01_S2-14 % kill -9 $(lsof -ti :5000) (venv) personal@Raguls-MacBook-Air Sprint2_S2-01_S2-14 % python app.py * Serving Flask app 'app' | Debug mode: on | Running on http://127.0.0.1:5000
```

S2-01: Take Quiz & Answer Selection

Module: Ragul | TC-04 / S2-01

The candidate interface for *Java Basics Quiz* loads five active multiple-choice questions. Radio buttons enforce mutually exclusive single selections per question, while a dynamic client-side timer counts down from 10:00 (09:40 elapsed) before triggering submission via POST `/quiz/1/submit`.

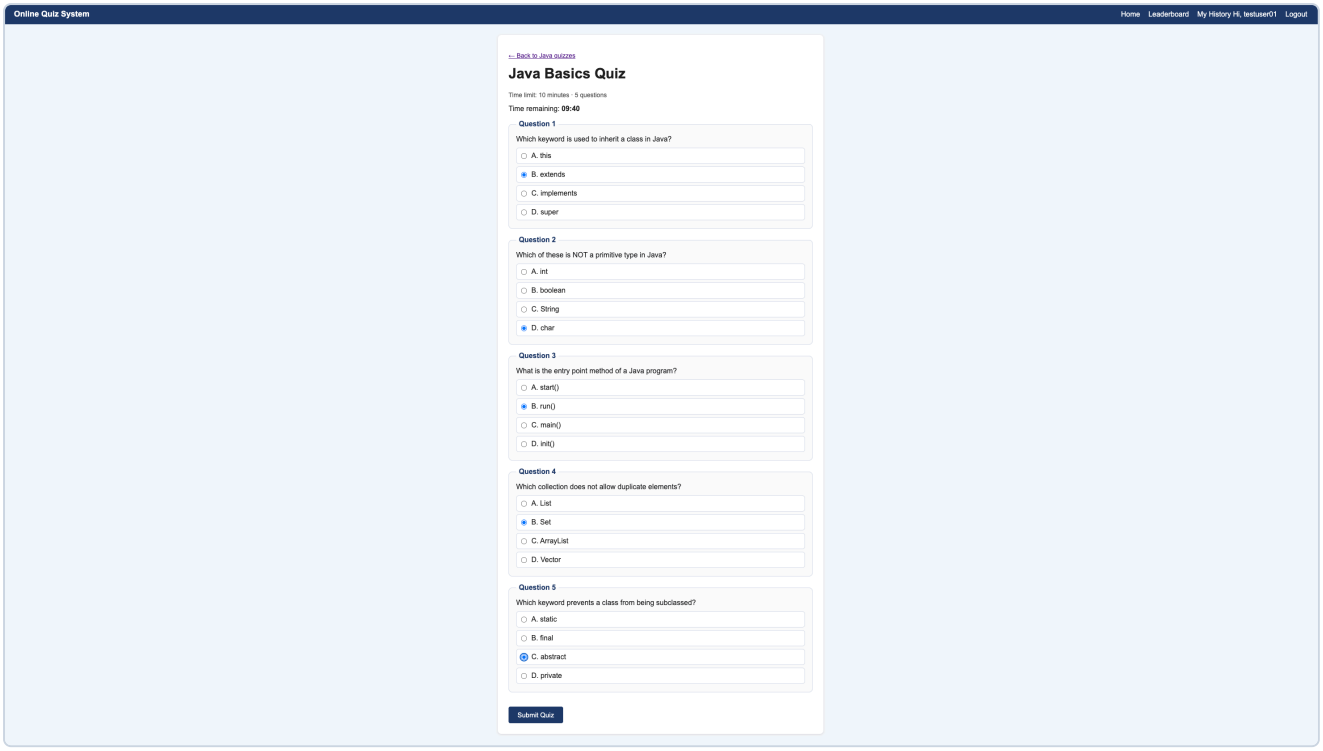


Figure 2.1: Active assessment interface showing question selections, timer (09:40), and the Submit Quiz button.

```
127.0.0.1 - - [29/Oct/2026 14:21:29] "GET /quiz/1 HTTP/1.1" 200 - Form State: Question 1 [Option B], Question 2 [Option D], Question 3 [Option B], Question 4 [Option B], Question 5 [Option C]
```

S2-04: Quiz Result Report (Scorecard & Review)

Module: Ragul | TC-13 / S2-04

Upon submission, the route `/result/<result_id>` grades candidate choices against database answer keys. The generated scorecard outputs Candidate Name (`testuser01`), Quiz Title, Category, Score (2/5), Time Taken (22s), timestamp, and a question-by-question review highlighting correct versus wrong choices.

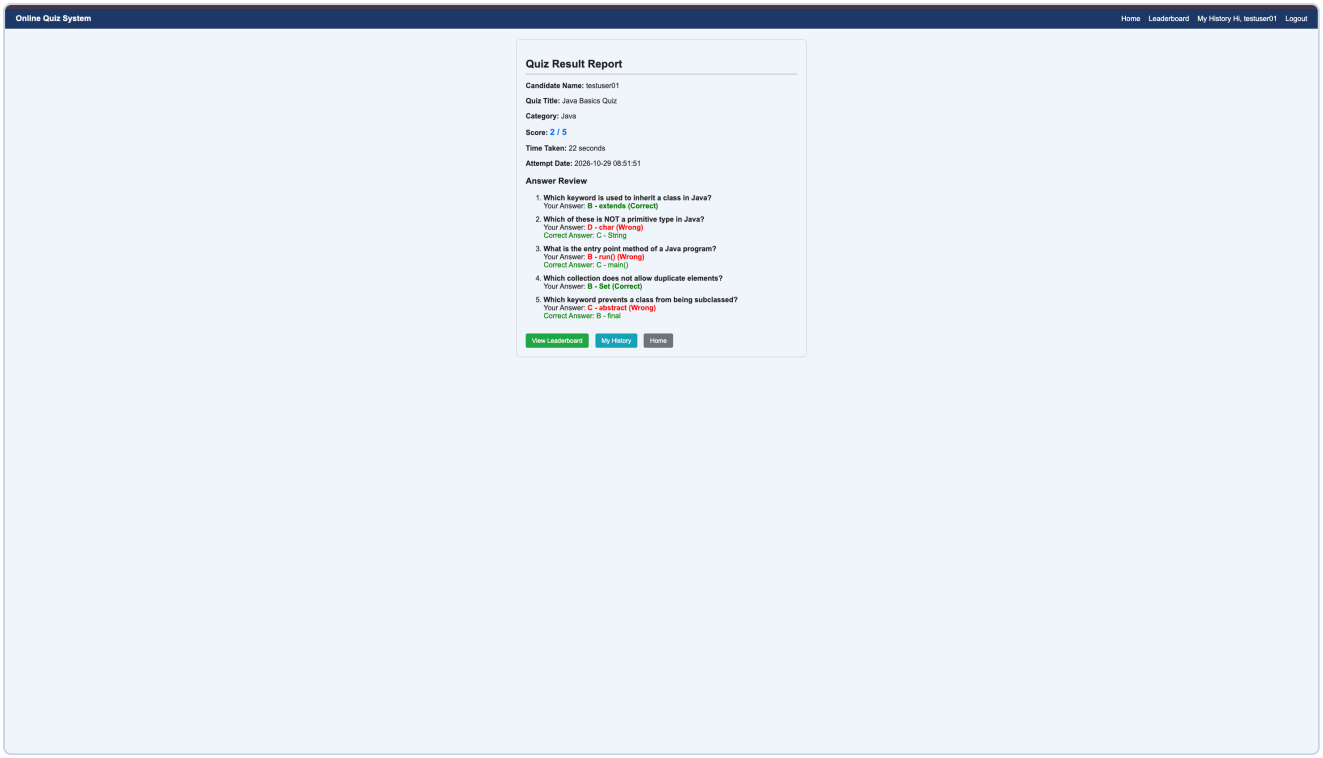


Figure 3.1: Candidate scorecard displaying score 2/5, completion duration 22 seconds, and question review.

```
127.0.0.1 - - [29/Oct/2026 14:21:51] "POST /quiz/1/submit HTTP/1.1" 302 - 127.0.0.1 - - [29/Oct/2026 14:21:51] "GET /result/2 HTTP/1.1" 200 - Result record committed: User ID 1 | Quiz ID 1 | Score 2 | Total 5 | Time 22s
```

S2-02 & S2-03: Real-Time Leaderboard & Tie-Breakers

Module: Ragul | TC-14 & TC-15

The leaderboard query sorts participants by highest score descending (`score DESC`). In the event of equal scores, a deterministic tie-breaker ranks the candidate with the fastest completion duration first (`time_taken ASC`). The dropdown selector filters rankings dynamically across quizzes.

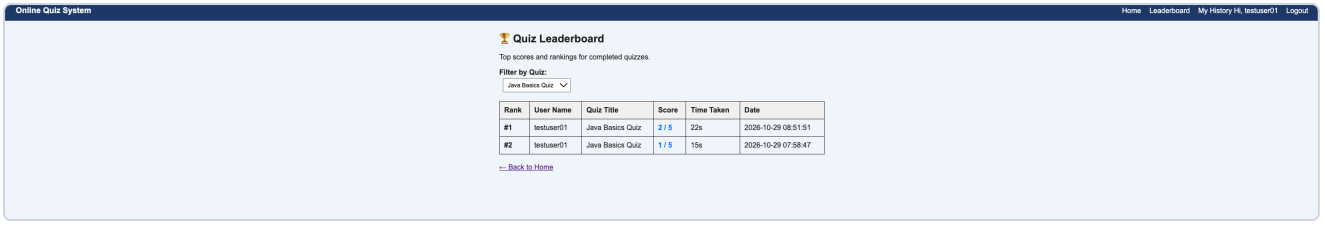


Figure 4.1: Quiz Leaderboard ranking testuser01 at #1 (2/5 in 22s) above prior attempt at #2 (1/5 in 15s).

```
127.0.0.1 - - [29/Oct/2026 14:22:04] "GET /leaderboard?quiz_id=1 HTTP/1.1" 200 - Query:
SELECT u.name, r.score, r.time_taken, r.submitted_at FROM Result r ORDER BY score DESC, time_taken
ASC;
```

TC-16: Candidate Quiz History & Result Persistence

Module: Ragul | TC-16 / S2-05

The route `/results/history` verifies that candidate submissions remain permanently stored in the SQLite database across sessions. The candidate views their chronological attempt logs and accesses individual scorecard reports directly via the "View Report" action link.

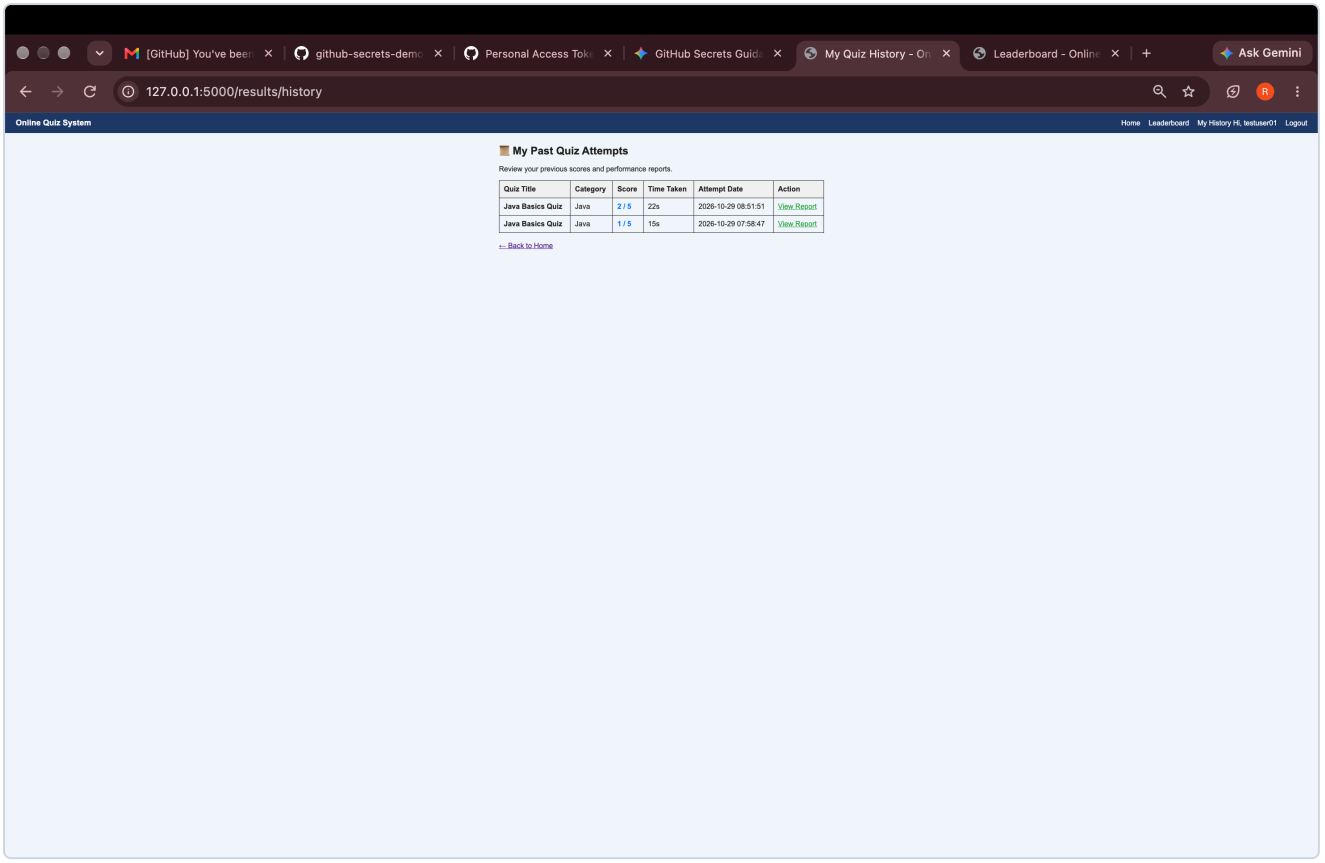


Figure 5.1: Candidate attempt history table detailing quiz title, category, score, duration, and report access.

```
127.0.0.1 - - [29/Oct/2026 14:22:54] "GET /results/history HTTP/1.1" 200 - Retrieved 2 historic attempts for user_id = 1: [Attempt #2: 2/5 (22s), Attempt #1: 1/5 (15s)]
```

TC-18: Role-Based Access Control (RBAC Protection)

Security & Authorization | TC-18

Role-based access control enforces administrative privileges using the custom `@admin_required` route decorator. When logged in as standard candidate `testuser01`, manual navigation to `/admin` is intercepted, returning a 302 redirect to `/home` with an alert banner.

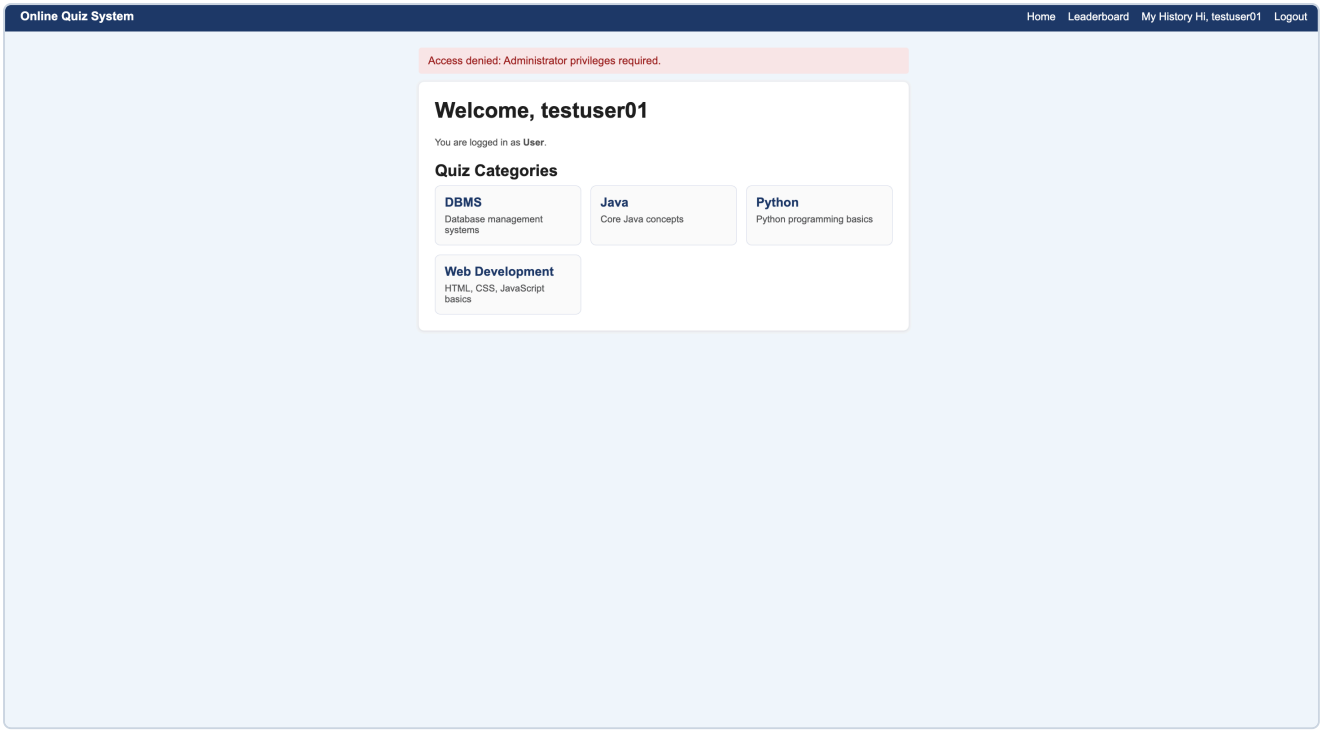


Figure 6.1: Candidate home view displaying access rejection banner: "Access denied: Administrator privileges required."

```
127.0.0.1 - - [29/Oct/2026 14:23:28] "GET /admin HTTP/1.1" 302 - 127.0.0.1 - - [29/Oct/2026 14:23:28] "GET /home HTTP/1.1" 200 - Security Event: Unauthorized role 'user' rejected from administrative controller.
```

S2-06: Administrator Dashboard Access & Navigation

Module: Sharanya | S2-06

Authenticating with administrator credentials (`admin01@example.com`) updates the session role to `admin`. The application navigation bar dynamically displays the yellow **Admin Panel** hyperlink, enabling administrative workflow access.

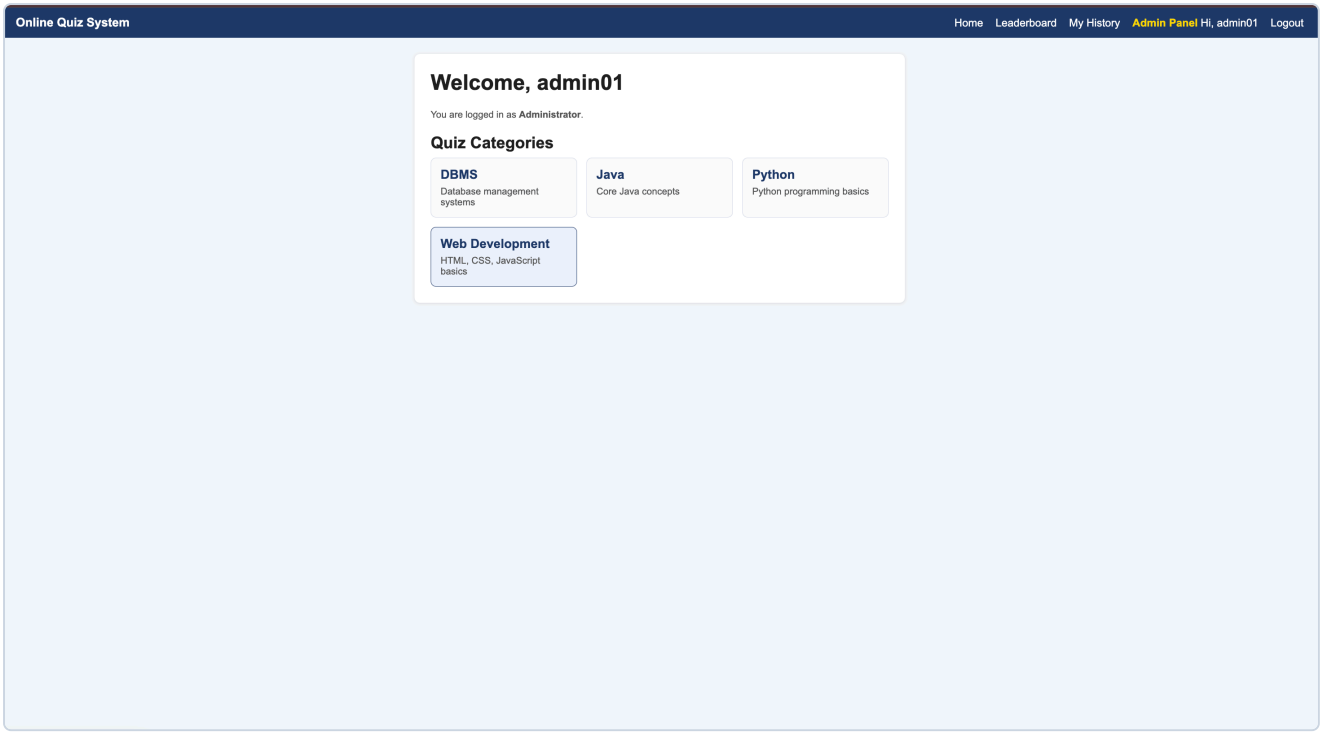


Figure 7.1: Navigation header rendered for admin01 showing the highlighted yellow Admin Panel control.

```
127.0.0.1 - - [29/Oct/2026 14:24:32] "POST /login HTTP/1.1" 302 - 127.0.0.1 - - [29/Oct/2026 14:24:32] "GET /home HTTP/1.1" 200 - Session Initialized: user_id = 2, email = 'admin01@example.com', role = 'admin'
```

S2-11: Add Quiz Category to Database

Module: Sharanya | TC-24 & S2-11

The Administrator Dashboard (`/admin`) features a dedicated form for adding quiz categories. Submitting *Cloud Computing* with description *AWS, Azure, and cloud infrastructure fundamentals* writes directly to SQLite and updates the category list in real time.

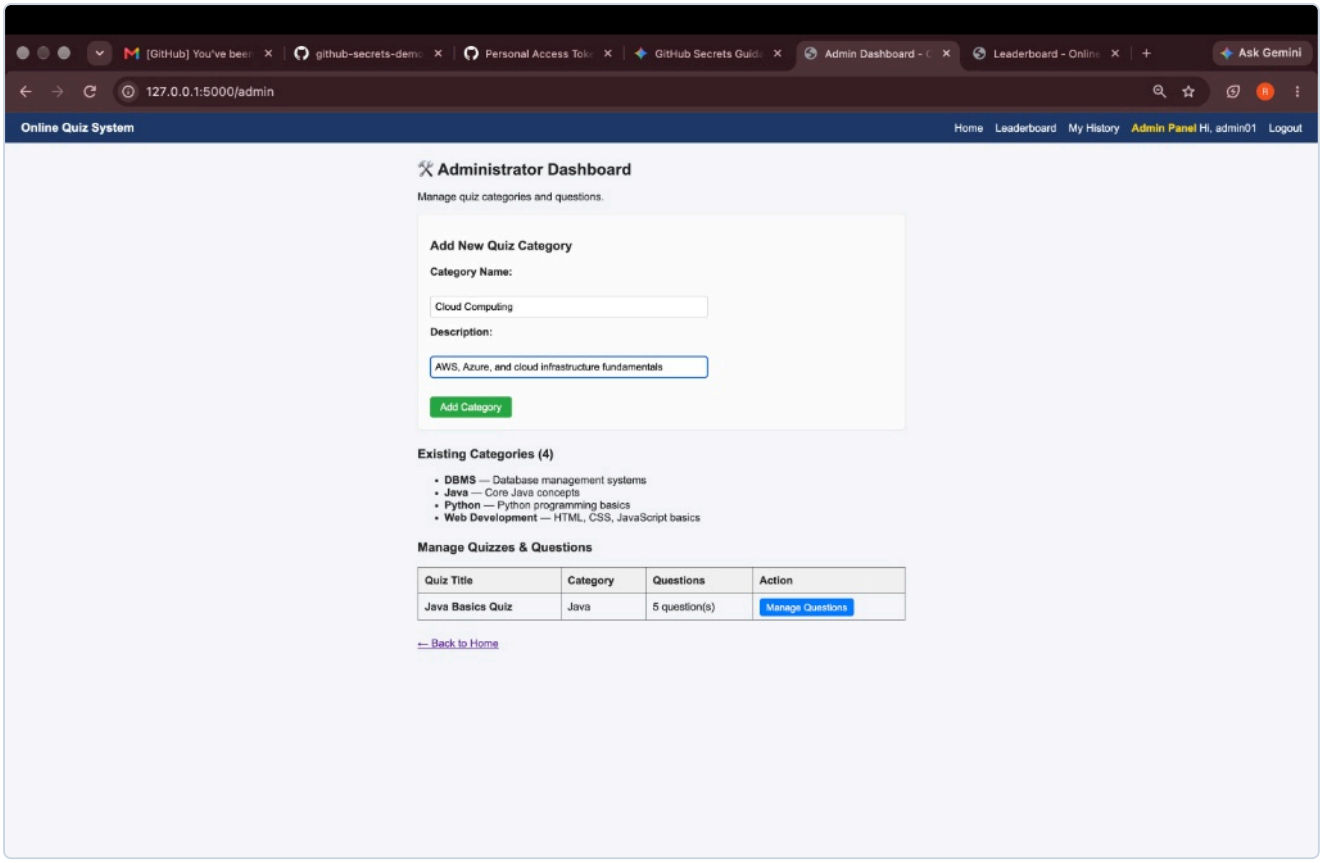


Figure 8.1: Administrator Dashboard with Category creation form and existing categories list.

```
127.0.0.1 - - [29/Oct/2026 14:26:28] "POST /admin/category/add HTTP/1.1" 302 - 127.0.0.1 - - [29/Oct/2026 14:26:28] "GET /admin HTTP/1.1" 200 - INSERT INTO Category (name, description) VALUES ('Cloud Computing', 'AWS, Azure, and cloud...');
```

S2-08: Add Question with Multi-Option Validation

Module: Sharanya | TC-19, TC-20, TC-21

In *Manage Questions* (`/admin/quiz/1/questions`), the administrator creates a new question by specifying the text, four distinct options (A through D), and the designated correct answer key (`Option B`). The item is validated and appended as Question 6.

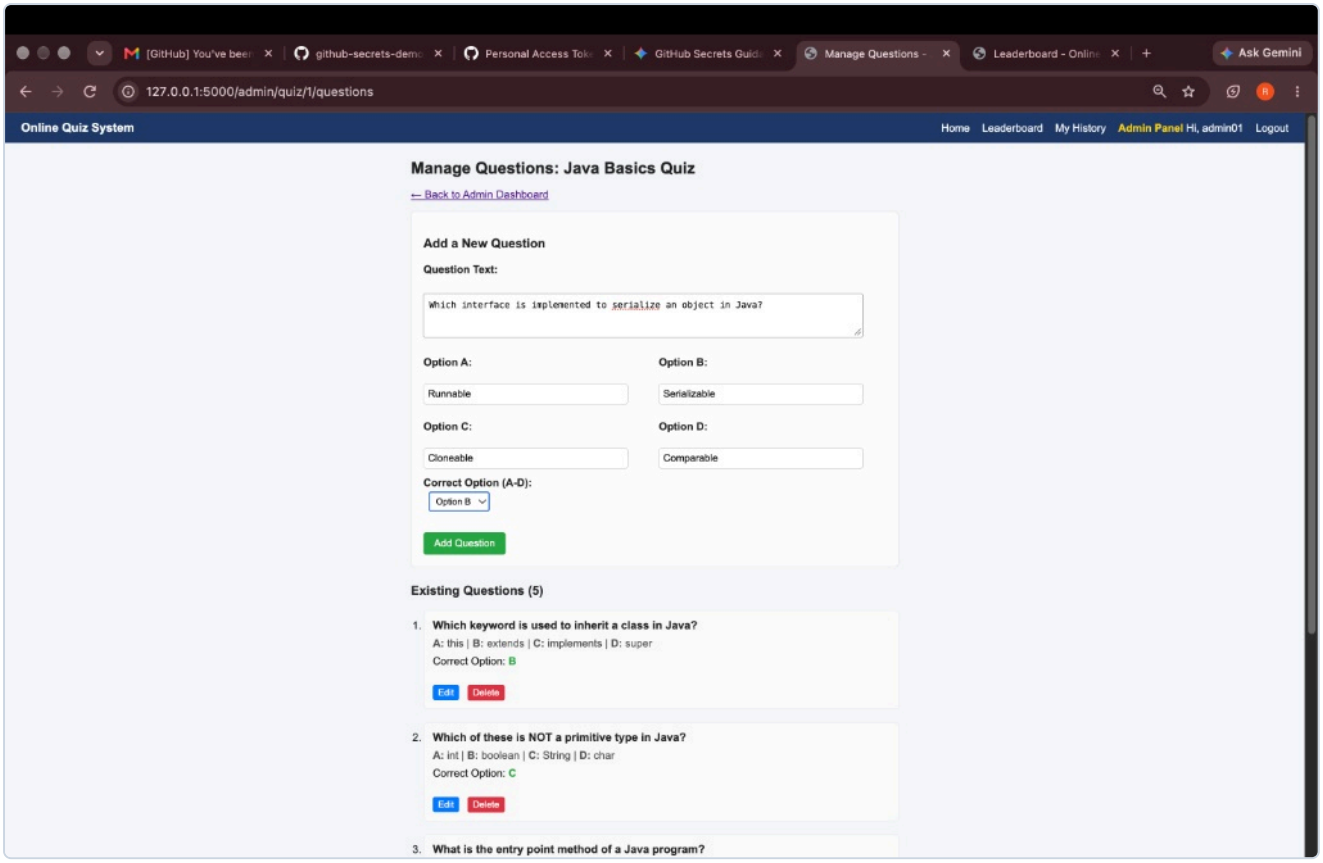


Figure 9.1: Question authoring form showing question description, options A–D, and correct key selection.

```
127.0.0.1 - - [29/Oct/2026 14:28:12] "POST /admin/quiz/1/question/add HTTP/1.1" 302 -
127.0.0.1 - - [29/Oct/2026 14:28:12] "GET /admin/quiz/1/questions HTTP/1.1" 200 - Question Added:
'Which interface is implemented to serialize an object in Java?' [Key: B]
```

S2-09: Edit Question Form Interface

Module: Sharanya | TC-22 / S2-09

Selecting the **Edit** button on Question 8 routes to `/admin/question/8/edit`. The form pre-populates existing record values, enabling text modifications, option alterations, and key reassignments before persisting updates back to SQLite.

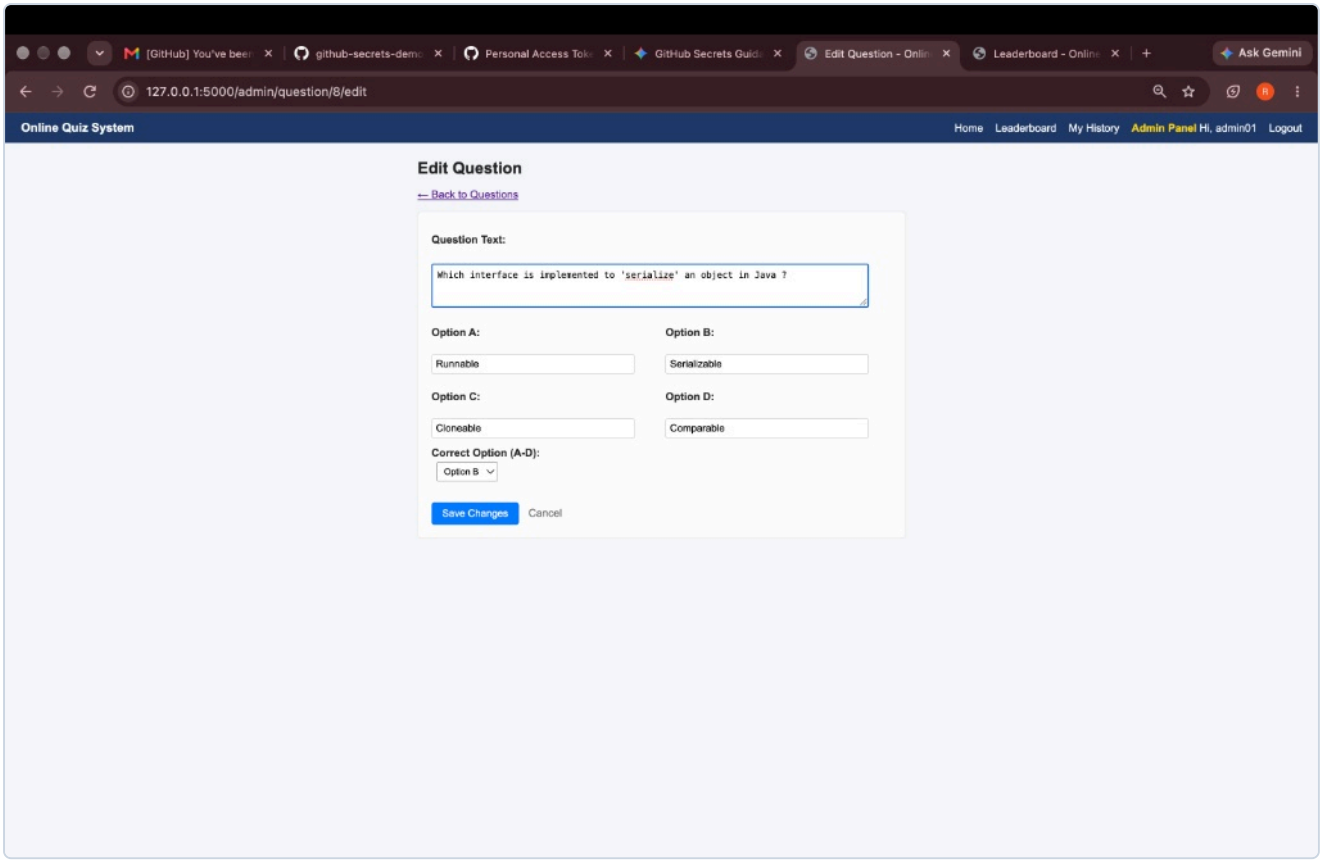


Figure 10.1: Edit Question interface pre-loaded with Question #8 text, options A–D, and correct key.

```
127.0.0.1 - - [29/Oct/2026 14:29:31] "GET /admin/question/8/edit HTTP/1.1" 200 - Editing Question ID 8: Text updated to 'Which interface is implemented to \'serialize\' an object in Java ?'
```

S2-09 (Cont.): Question Update Confirmation Banner

Module: Sharanya | TC-22 / S2-09

After clicking **Save Changes**, the route `POST /admin/question/8/edit` executes an UPDATE statement. The client is redirected back to the questions list with a green confirmation banner: *"Question updated successfully."*

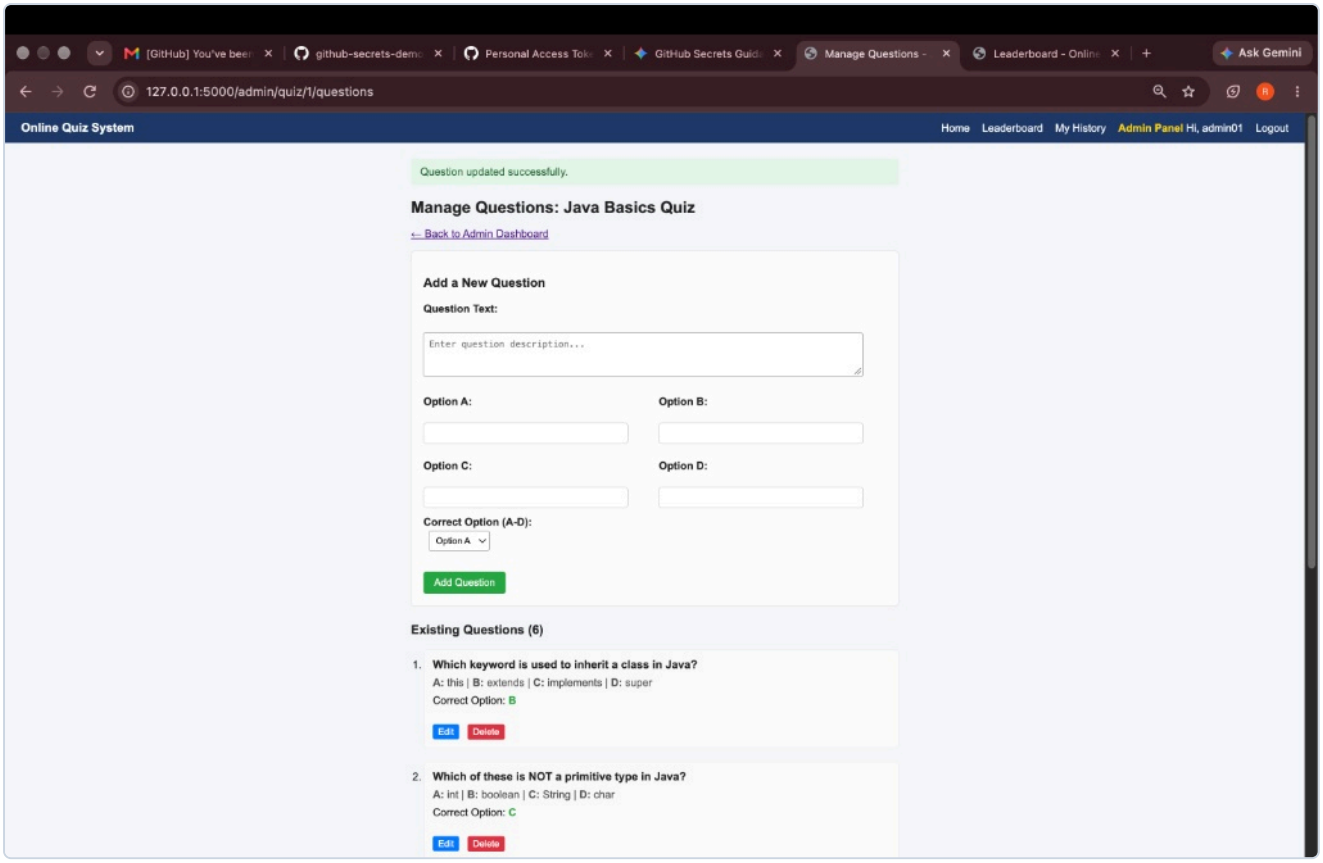


Figure 11.1: Question management view displaying the success alert banner and 6 active questions.

```
127.0.0.1 - - [29/Oct/2026 14:29:59] "POST /admin/question/8/edit HTTP/1.1" 302 - 127.0.0.1 - - [29/Oct/2026 14:29:59] "GET /admin/quiz/1/questions HTTP/1.1" 200 - Database update verified: Row updated in table 'Question' for question_id = 8.
```

S2-10: Delete Question & Confirmation Dialog

Module: Sharanya | TC-22 / S2-10

Clicking **Delete** prompts an explicit browser confirmation dialog (*"127.0.0.1:5000 says: Are you sure you want to delete this question?"*) to prevent accidental deletions. Confirming issues a deletion request and returns the count to five.

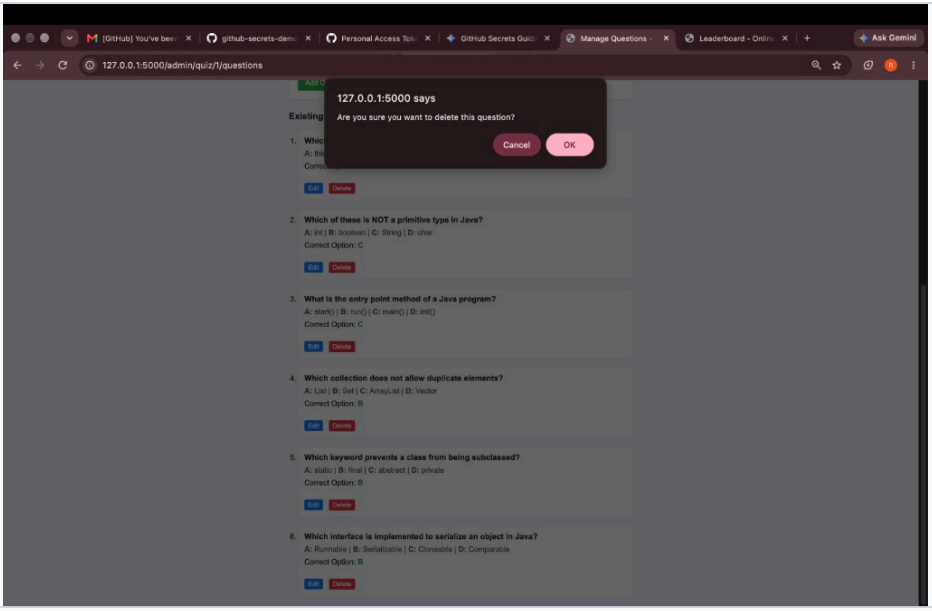


Figure 12.1: Deletion confirmation modal preventing unintentional data loss before execution.

Module ID	Feature / User Story	Responsible	Verification Status
S2-01 to S2-05	Timer, Scoring, Leaderboard, History, RBAC	Ragul	PASSED
S2-06 to S2-11	Admin Suite, Categories, Question CRUD	Sharanya	PASSED

```
127.0.0.1 - - [29/Oct/2026 14:30:31] "POST /admin/question/8/delete HTTP/1.1" 302 - 127.0.0.1 - - [29/Oct/2026 14:30:31] "GET /admin/quiz/1/questions HTTP/1.1" 200 - Active Questions in 'Java Basics Quiz': 5 (Verification complete)
```